



Speaker Identification from Movie-Watching sEEG Data

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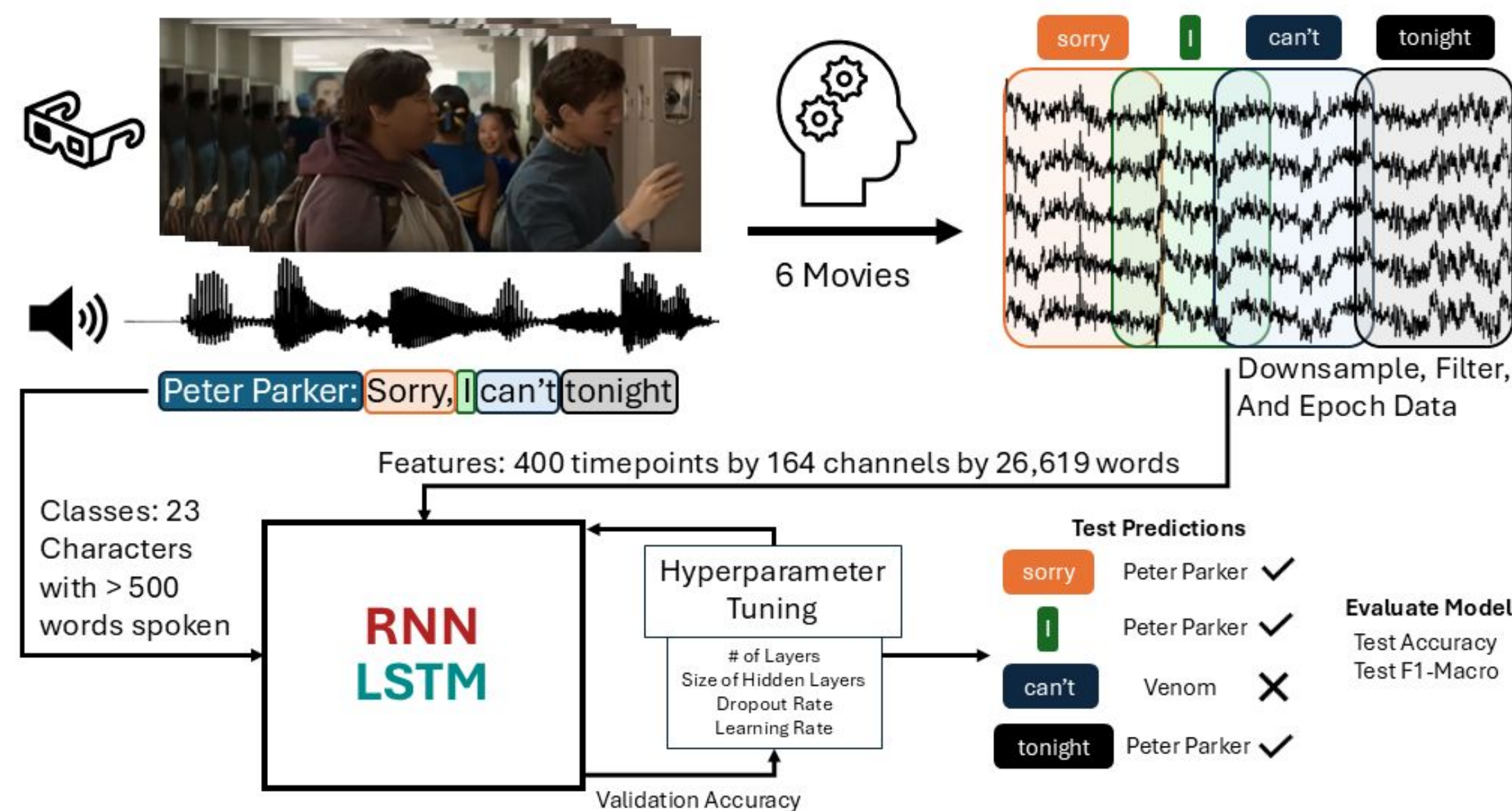
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CS229: Machine Learning

Background

A key research area in BCIs focuses on **decoding neural activity** in **naturalistic settings**, which better reflects real-world experiences compared to controlled experiments. This project addresses this challenge using the Brain Treebank repository,¹ analyzing 13.5 hours of movie-watching sEEG data with **RNNs** and **LSTMs** to **classify the current speaker**. By improving neural decoding methods, this research aims to enhance real-time neural interpretation in immersive environments and other communication tools.

Methods



Data and Feature Extraction

Data: Subject 2 from the Brain Treebank.¹ 13.5 hours of movie-watching neural data across 6 movies, collected by **164 sEEG electrodes**. This resulted in **28,619 spoken words** with **23 annotated speakers**.

Feature Extraction: sEEG data was filtered and downsampled to 200 Hz then epoched +/- 1 sec after word onset to form samples. Reduced dataset to **speakers with more than 500 words** spoken for sufficient training data. Stratify split into 80/10/10 train/validation/test splits with balanced classes.

Model Architecture

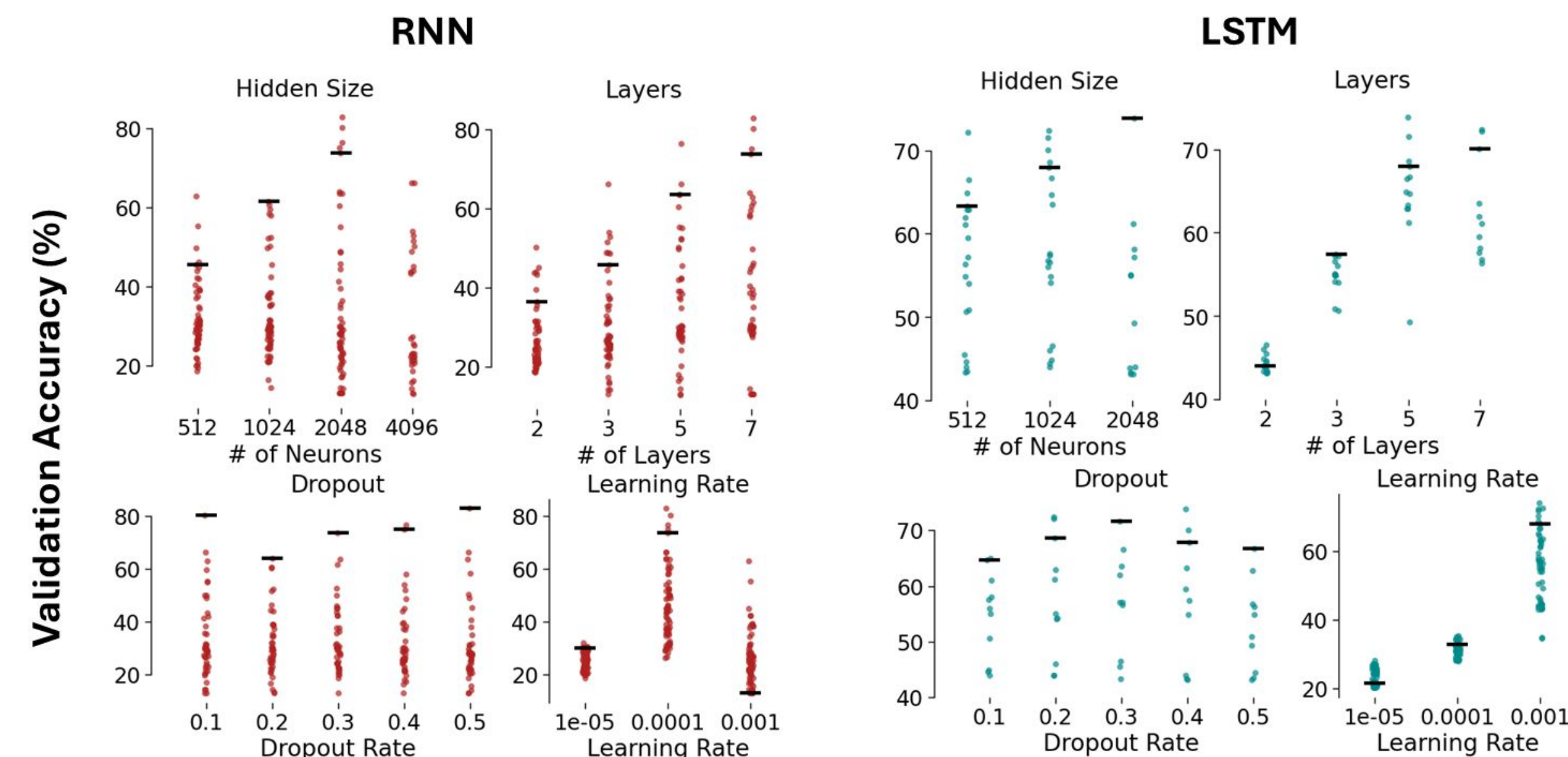
We implemented two neural network based models for this task.

- **Deep RNN:** Our model is a multi-layer RNN with a tanh activation function with dropout applied to the inner layers, a final dropout layer, then a fully-connected layer with a cross-entropy loss function.
- **LSTM:** LSTMs are used to mitigate the vanishing gradient problem associated with RNNs. Our LSTM neural network follows the same design as above, with the addition of a memory cell and input, output, and forget gates to determine what information is stored and passed on.

Experiments

Hypothesis: We hypothesized that the **LSTM architecture** would **outperform traditional RNN models**. Similarly, we also hypothesized that **increasing hidden size and number of layers** would **significantly increase model performance**.

Hyperparameter Tuning



Note: We spent 7 and 11 days respectively training these RNN and LSTM models on a Nvidia RTX 3090 before running out of compute resources and time to further hyperparameter tune. We set a max epoch limit of 100.

Hyperparameter Tuning: Grid search on hidden layer size, number of layers, dropout rate, and learning rate.

Key Findings: Optimal learning rates were 10^{-4} (RNN) and 10^{-3} (LSTM); increasing neurons and layers improved performance, while dropout, though not significantly sensitive to rate, consistently enhanced results compared to no dropout (which remained near baseline).

Model Results

Best RNN Models (LR = 10^{-4})

Hidden	Layers	Dropout	Val	Test
2048	7	0.5	82.8	80.5
2048	7	0.1	80.3	78.7
2048	5	0.4	76.5	74.3
2048	7	0.4	75.1	72.7
2048	7	0.3	73.7	71.8

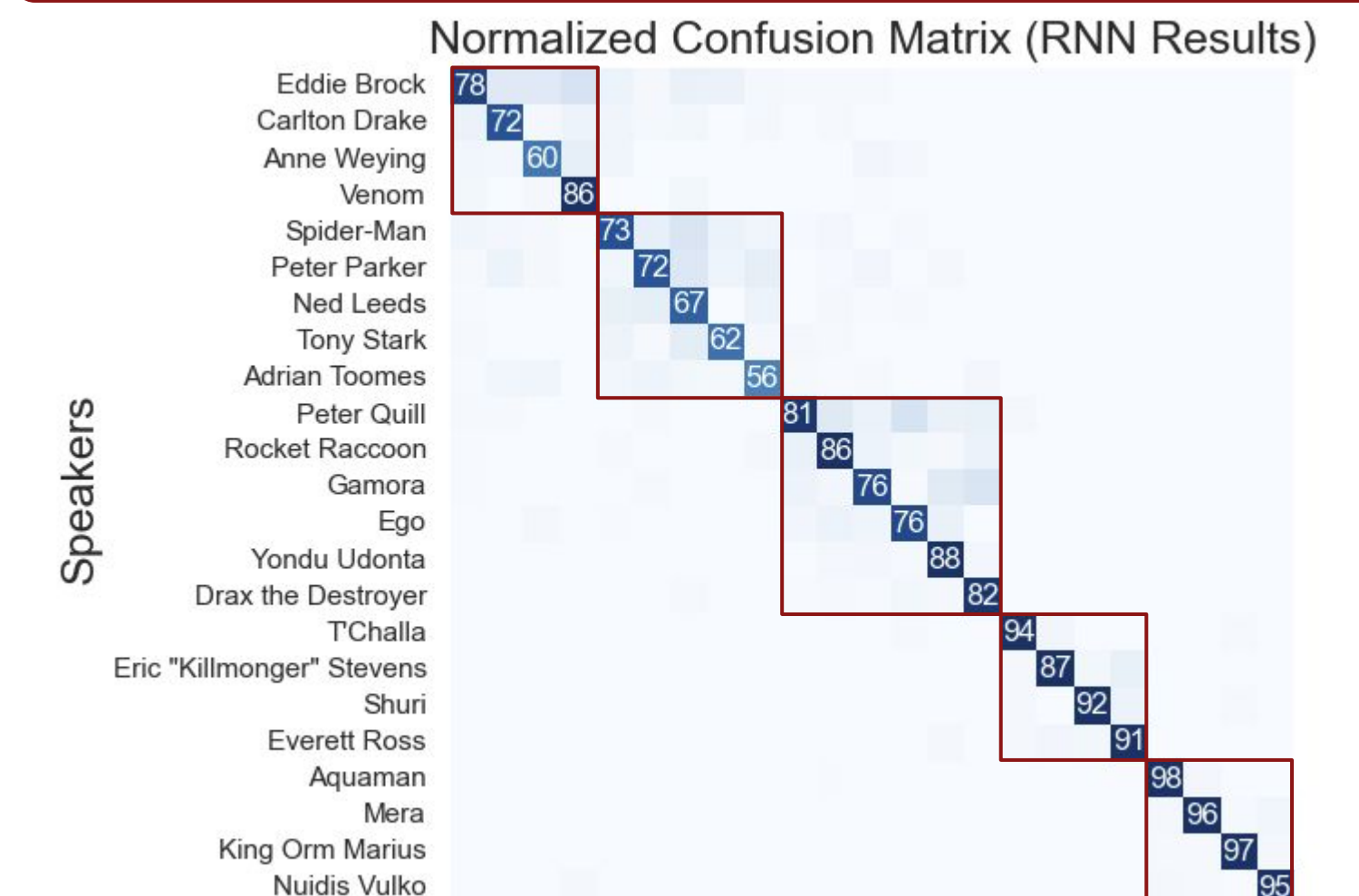
Best LSTM Models (LR = 10^{-3})

Hidden	Layers	Dropout	Val	Test
2048	5	0.4	73.9	72.8
1024	7	0.2	72.4	71.3
512	7	0.2	72.2	71.3
1024	5	0.3	71.6	71.2
1024	7	0.1	70.1	67.7

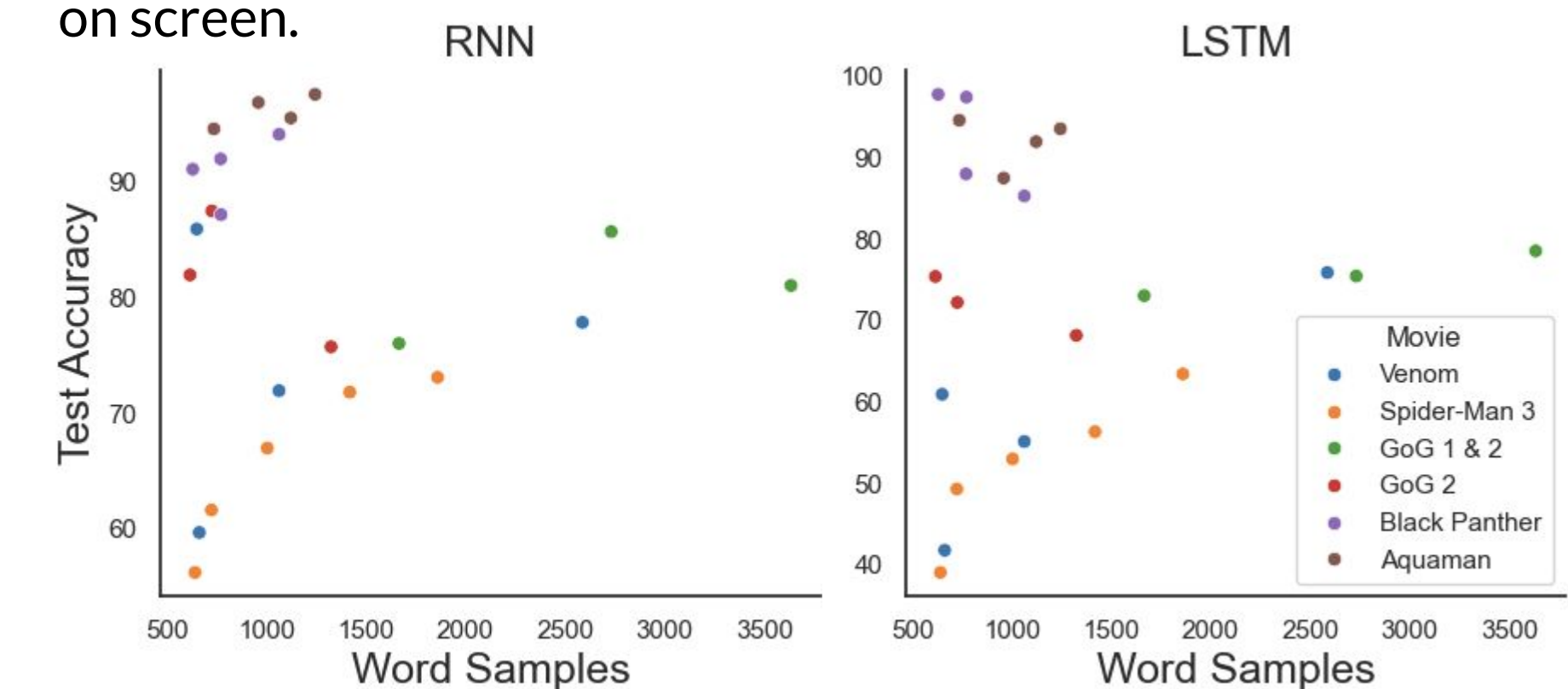
Model Comparison: Our best RNN model has a test accuracy of 80.5% and our best LSTM model has a test accuracy of 72.8%. This is significantly above the naive MLE baseline of 9.1% (found by taking the largest class and dividing by the total number of samples).

Thus our results contradict our first hypothesis, with **RNN models actually performing better than LSTM models**. However our **second hypothesis holds with increased layers and hidden size correlating with improved performance** across all models.

Discussion and Analysis



Intramovie Misclassification: Our models misclassify characters within the **same movie** at a **higher rate** than out-of-group speakers. This indicates that the models are picking up on features inherent to each movie such as **environmental cues** or the visual stimuli of **multiple characters** on screen.



Class Imbalances: We found that per-class test accuracy seems to be a function of the **number of samples** and **intramovie class balance**. Movies with a smaller standard deviation like Black Panther or Aquaman (~200 words) had an accuracy >90% despite lower samples on average. Movies with a larger standard deviation (500-900) had a lower average accuracy of ~70% with the relationship between per-class accuracy and number of samples being **roughly logarithmic**.

Future Work

In the future we'd like to:

- Use **Weighted Cross-Entropy Loss** to mitigate class imbalances.
- **Evaluate Larger Models** based on results of our hyperparameter tuning.
- Evaluate **other subjects with less data** to test robustness.

¹⁾ C. Wang, A. U. Yaari, A. K. Singh, V. Subramaniam, D. Rosenfarb, J. DeWitt, P. Misra, J. R. Madsen, S. Stone, G. Kreiman, B. Katz, I. Cases, and A. Barbu, "Brain Treebank: Large-scale intracranial recordings from naturalistic language stimuli," *arXiv preprint arXiv:2411.08343*, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2411.08343>